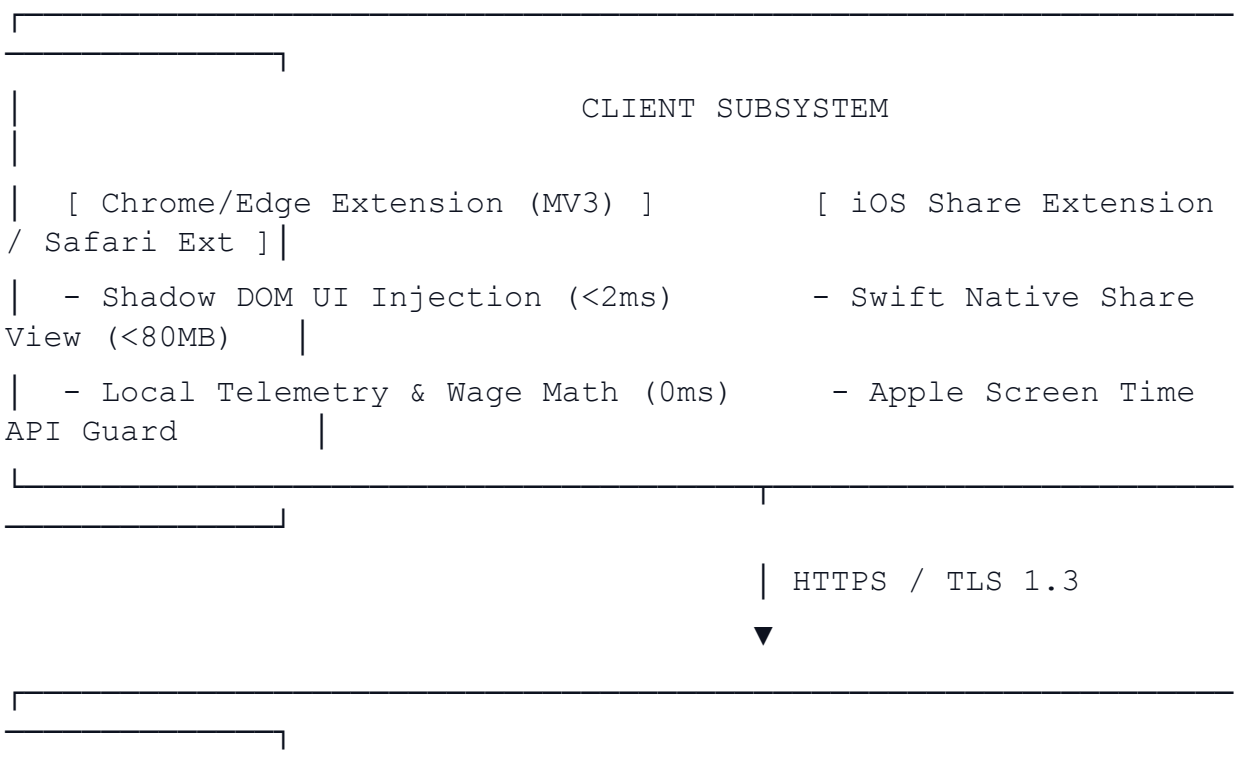
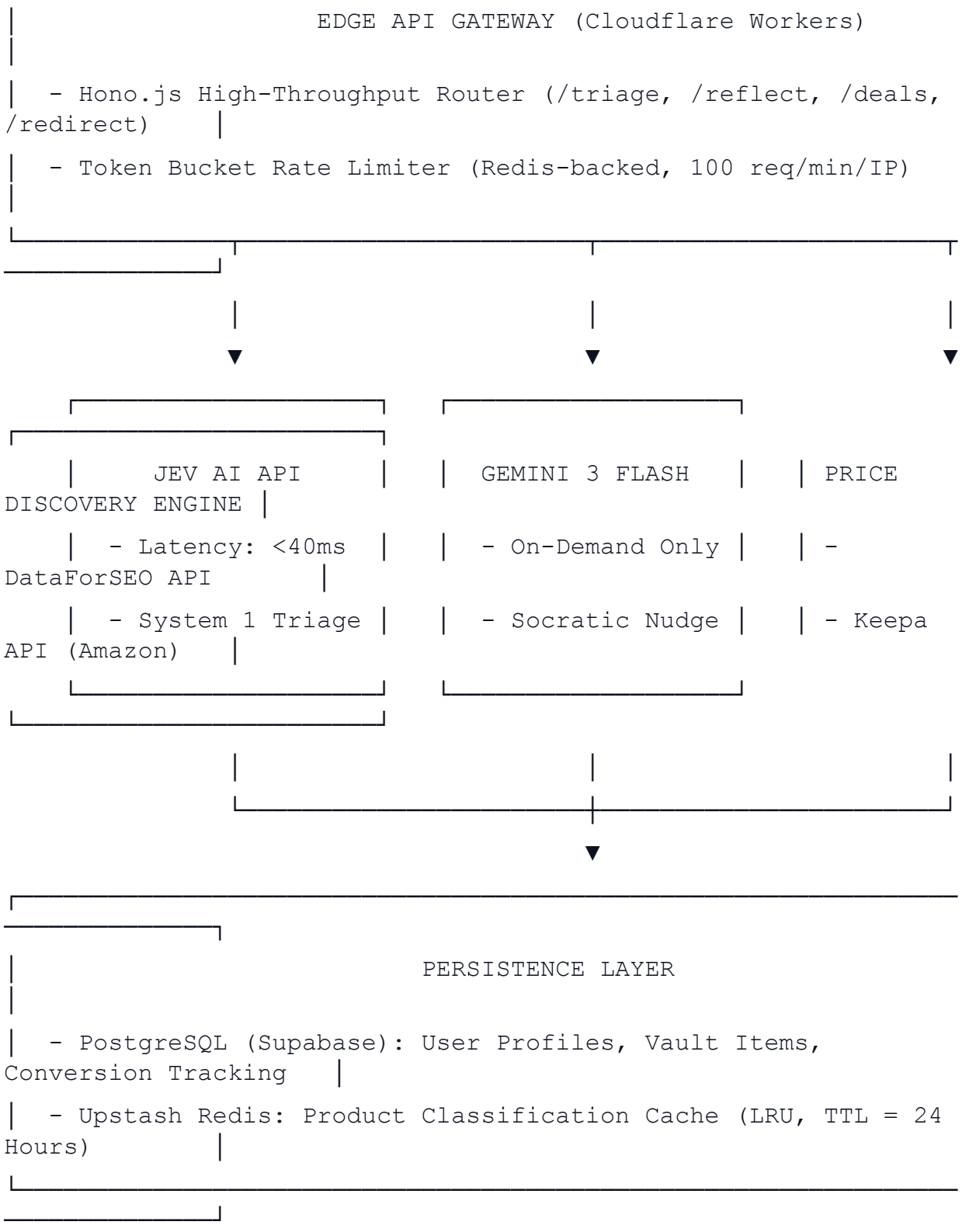
ZENBUY: FINAL TECHNICAL PRODUCT SPECIFICATION

Mathematical Impulse Risk Engine, Algorithmic Weights, AI Contracts & System Topology | Version 1.0 Production Baseline

1. System Architecture Overview & Topological Invariants

ZenBuy is an ambient, dual-process client-server system designed to detect, evaluate, and mediate consumer e-commerce purchase impulses in real time. It prioritizes client-side zero-knowledge privacy: personal financial figures (e.g., hourly wage, budgets) remain strictly in local browser storage, while anonymized product metadata is triaged via low-latency edge compute.





2. The Mathematical Impulse Risk Scoring Engine (IRS)

The core decision engine evaluates a composite **Impulse Risk Score (IRS)** normalized between `0` and `100` upon every user purchase interaction:

$$\mathbf{IRS} = \sum_{i=1..5} (w_i \times S_i) \in [0, 100]$$

$$\text{IRS} = (0.35 \times S_{\text{Taxonomy}}) + (0.20 \times S_{\text{Velocity}}) + (0.15 \times S_{\text{Circadian}}) + (0.15 \times S_{\text{DarkPattern}}) + (0.15 \times S_{\text{Financial}})$$

2.1 Dimension Weights & Parameter Scoring Formulas

Dimension	**Weight (w_i)**	**Sub-Score (S_i) Evaluation Method**	**Telemetry / Data Source**
1. Product Taxonomy	**0.35 (35%)**	Hedonic vs. Utilitarian classification index (0 to 100).	Jev AI sub-50ms structured JSON classification.
2. Shopping Velocity	**0.20 (20%)**	Dwell time, traffic source, scroll depth, comparison tabs.	Client-side `performance.timing` & document referrer.
3. Circadian State	**0.15 (15%)**	Ego-depletion factor based on local client timestamp.	Local device clock (peak risk 11:00 PM – 4:59 AM).
4. Dark Pattern Exposure	**0.15 (15%)**	Mathur et al. (Princeton) deceptive interface score.	DOM scanner (fake timers, social proof toasts, pre-checks).
5. Financial Salience	**0.15 (15%)**	Labor-to-price ratio and BNPL installment usage.	

Item price vs. client local wage and payment selection.

2.2 Detailed Sub-Score Parameter Matrices

Dimension 1: Product Taxonomy & Hedonic Index (S_1, Weight = 0.35)

- **Household Staples / Medical (0 – 15 pts):** Baby formula, diapers, groceries, prescription refills, lightbulbs.
- **Utilitarian / Tools / Work (20 – 40 pts):** Cables, replacement winter boots, textbooks, phone repair parts.
- **Discretionary Wants (50 – 75 pts):** Board games, casual sneakers, headphones, kitchen gadget upgrades.
- **Hedonic Luxury / Indulgence (76 – 90 pts):** Designer accessories, perfumes, mechanical keyboards, gaming skins.
- **Viral Novelties / Fads (91 – 100 pts):** TikTok-viral dropshipped items, novelty ambient lamps, gadget trinkets.

Dimension 2: Browsing Velocity & Context (S_2, Weight = 0.20)

- **Social Media Ad Referrer (+35 pts):** `document.referrer` matches TikTok, Instagram, Pinterest, Facebook.
- **Direct Search Intent (-30 pts):** Arrived via Google search query matching specific product model/part number.
- **Dwell Time < 30 seconds (+25 pts):** Cart click occurred under 30s from page load (impulse trigger).
- **Dwell Time > 4 minutes (-20 pts):** Prolonged inspection of specifications and technical details.
- **Deep Review Reading (-15 pts):** User lingered > 45s on 1-star/3-star critical reviews.
- **Multi-Tab Comparison (-15 pts):** User has 3+ competing product tabs open simultaneously.

Dimension 3: Circadian State & Ego Depletion (S_3, Weight = 0.15)

Based on Loewenstein's visceral states and Kahneman's Dual-Process cognitive depletion:

```
Local Hour (H in [0..23]):
- If H in [23, 0, 1, 2, 3, 4] (11:00 PM - 4:59 AM) ➔ S3 = 100 (Peak Ego Depletion)
- If H in [21, 22] or H == 5   (9:00 PM - 10:59 PM) ➔ S3 = 60 (Late Evening Fatigue)
- If H in [18, 19, 20]         (6:00 PM - 8:59 PM)  ➔ S3 = 20 (Post-Work Transition)
- If H in [6..17]              (6:00 AM - 5:59 PM)  ➔ S3 = 0 (Daytime Rational Baseline)
```

Dimension 4: Mathur Dark Pattern Manipulation Index (S_4, Weight = 0.15)

Directly implementing the detection heuristics established by Mathur et al. (Princeton 2019) and Luguri & Strahilevitz (2021):

- **Deceptive Urgency (+40 pts):** Timer detected in DOM that resets upon page reload (verified via `sessionStorage` audit).
- **Fabricated Social Proof (+25 pts):** Injected toast notifications from dark pattern SaaS scripts (`usefomo`, `fera.ai`, `beeketing`).
- **Low-Stock Ambiguity (+20 pts):** "Only 2 left in stock!" message unaccompanied by live inventory metadata.
- **Sneak-into-Cart Detected (+35 pts):** Line-item diff reveals auto-added package protection or warranty add-ons.
- **Preselection / Forced Continuity (+25 pts):** Pre-checked "Subscribe & Save" or recurring charge defaults.

Dimension 5: Financial Salience & Payment Decoupling (S_5, Weight = 0.15)

Labor Ratio `R = Price / Hourly_Wage`:

- **Labor Ratio R ≥ 5.0 hours (+40 pts):** Purchase represents more than half a working day.
- **Labor Ratio 2.0 ≤ R < 5.0 hours (+25 pts):** Purchase represents 2 to 5 hours of work.
- **Labor Ratio R < 1.0 hour (0 pts):** Purchase represents negligible labor investment.
- **BNPL Installment Toggle Active (+30 pts):** User selected Klarna, Afterpay, or Affirm to disguise full cost.
- **Absolute Price ≥ $150 (+20 pts):** High-ticket impulse threshold.

3. Action Trigger Policies & Thresholds

Threshold Tier

Score Range

Action Policy

User Experience & Technical Execution

Tier 0: Silent Pass-Through

`IRS < 30`

`ACTION_SILENT`

Zero friction. Injected scripts remain inactive. Native "Buy Now" operates normally. (Staples, groceries, diapers).

Tier 1: Passive Choice Architecture

`30 ≤ IRS < 65`

`ACTION_PASSIVE_ICE`

Shadow DOM injects a sleek secondary **"🧊 Put on Ice (48h)"** button adjacent to Add to Cart. Price badge shows work hours. User decides freely.

Tier 2: Proactive Socratic Intercept

`IRS ≥ 65`

`ACTION_INTERCEPT`

Cart click is intercepted. ZenBuy modal opens displaying labor trade-off, Gemini Socratic copy, and 48-hour Vault freeze CTA.

4. Dual AI Performance & Execution Characteristics

4.1 Tier 1: Jev AI Triage Service (Sub-50ms SLA)

```
// TypeScript Contract for Jev AI Input Payload
export interface JevTriageRequest {
  productTitle: string;
  categoryPath?: string;
  priceUsd: number;
  localHour: number; // 0-23
  hasUrgencyBanner: boolean;
}

// Jev AI Output Judgment (Strictly-Typed JSON Schema)
export interface JevTriageResponse {
  classification: 'ESSENTIAL_STAPLE' | 'PRACTICAL_REPLACEMENT' |
'HEDONIC_WANT' | 'VIRAL_TREND';
  hedonicScore: number; // 0 to 100
  recommendedAction: 'SILENT_ALLOW' | 'PASSIVE_ICE_BUTTON' |
'PROACTIVE_INTERCEPT';
  confidence: number;   // 0.0 to 1.0
}
```

4.2 Tier 2: Gemini 3 Flash Socratic Engine

Invoked *strictly on-demand* when `IRS ≥ 65` or upon generating the 48-hour cooldown notification:

```
// Runtime Configuration
const geminiConfig = {
  model: 'gemini-2.5-flash',
  config: {
    temperature: 0.35,      // Controlled, non-hallucinatory
```

```
    topP: 0.85,
    maxOutputTokens: 120,    // Strictly 2 concise sentences
    systemInstruction: `You are ZenBuy, a calm, non-judgmental
financial alter-ego.
Reframe purchase desires into labor hours and long-term
financial freedom.
Never shame or scold. Acknowledge the desire, present the
trade-off, and recommend a 48h vault freeze.`
  }
};
```

5. Post-48h Deal Discovery & Affiliate Monetization Pipeline

1. **Universal Barcode Scraping:** Client extracts `gtin13`, `upc`, or `asin` from hidden `schema.org/Product` JSON-LD.
2. **Cross-Store Price Lookup:** Queries DataForSEO Google Shopping API across 20+ major US stores (Walmart, Target, Best Buy, eBay).
3. **Amazon Price Drop & Clip Coupon Scanner:** Queries Keepa API to inspect BuyBox status and historical lows.
4. **Affiliate Redirection Gateway:**
    - Amazon destinations: Direct Amazon Associates tag injection (`?tag=zenbuy-20`).
    - Other merchants: Routed via Skimlinks Universal Aggregator (covering 50,000+ stores with 3%–10% commission).

6. Client Footprint & Scaled Economics

6.1 Client Performance Budgets

- **Chrome Extension (MV3):** Minified bundle < 1.2 MB; runtime memory heap < 22 MB; DOM injection overhead < 3ms via Shadow DOM.
- **iOS Share Extension:** Compiled binary < 8.5 MB; memory footprint < 45 MB (well under Apple's 120MB limit); cold start < 250ms.

6.2 Monthly Scaled Unit Economics (Per Active User)

Operating Cost per Active User / Month: $0.029

Affiliate Revenue per Active User / Month: +$3.600 (4% on $90 intentional purchases)

Net Monthly Margin per Active User: +$3.571 (99.2% Gross Margin)

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